IVADO · ÉQUIALGO · CODEML 2026

Same R score, different postal code, different answer.

Finding and removing two hidden regional biases in scholarship decisions

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THE PROBLEM

An 88%-accurate model that copies a biased committee

48.4%

grant rate, Montréal and Capitale-Nationale

27.3%

grant rate, Bas-Saint-Laurent, Côte-Nord, Gaspésie

0.7

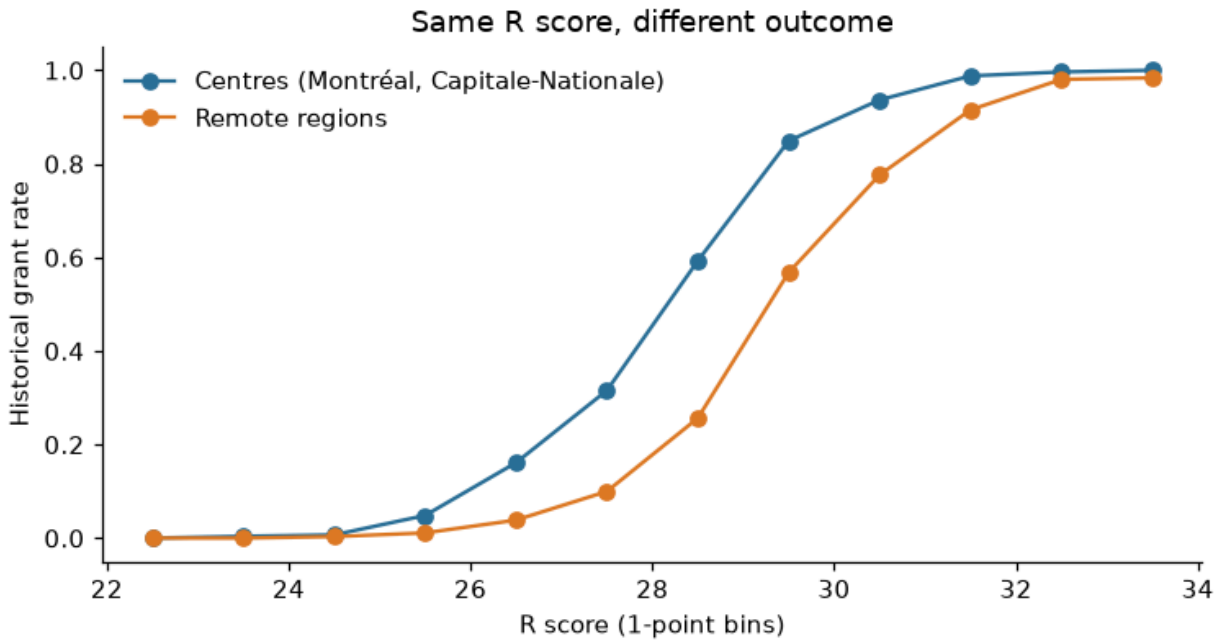
R-score points between the two groups, on a scale with SD 3

A 21-point gap in outcomes, a small gap in grades. The model is "accurate" because it reproduces past decisions, and those decisions are under audit.

Data: 10,000 historical files, 4,000 to score (synthetic)

BIAS CHANNEL 1

The committee docked remote applicants about 1.6 R points



- Between R 28 and 29: **59%** granted in centres, **26%** in remote regions.
- The committee applied a flat penalty of **≈ 1.6 R points** (95% CI 1.4–1.7), the same in all 3 remote regions. It explains **83%** of the gap.

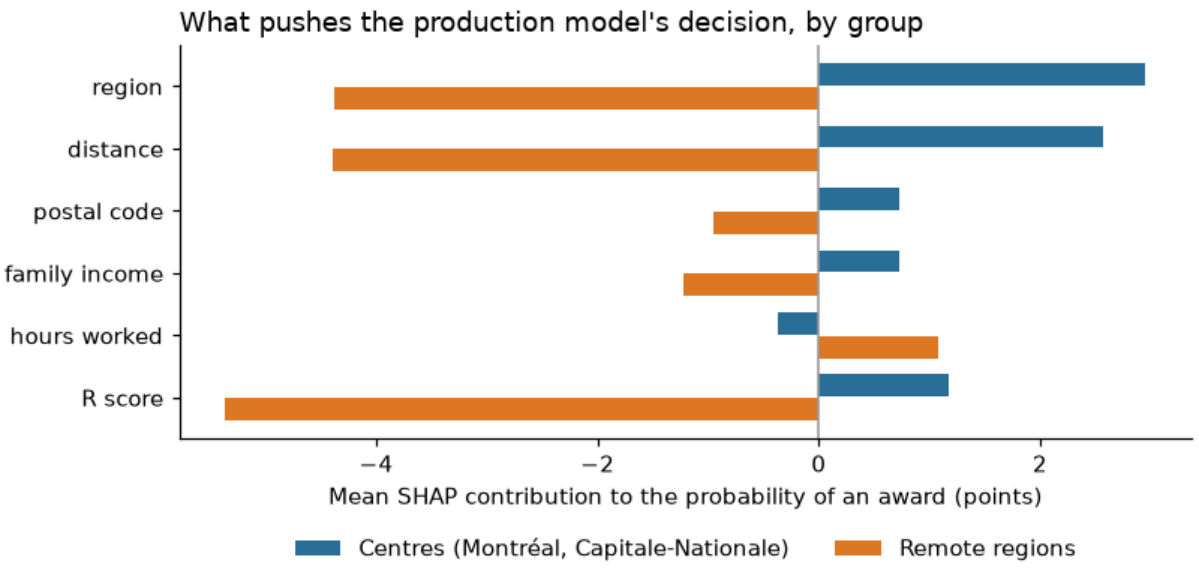
190 of 1,628

remote applicants the production model would fund if they simply lived in Montréal (same grades, hours, income)

audit_rapport.ipynb, sections 2–3

WHY "JUST DELETE THE COLUMN" FAILS

Proxies rebuild the penalty



- SHAP: **distance costs remote applicants as much as the region label**. Location moves their probability by **−10 points** (centres: +6).
- Distance and postal code predict region almost perfectly (AUC ≈ 1.0); hours 0.80; income 0.68.
- Delete the region column and the penalty moves onto distance.

The bias is in the labels, not the columns.

audit_rapport.ipynb, sections 4 and 5b

BIAS CHANNEL 2

The committee also rewarded family income

- Remote families earn **25% less** (median 51.6 k$ vs 69.2 k$). An income bonus works against them too.
- The two biases stack: **4 in 10 deserving low-income remote students were refused** (TPR 0.58).
- We tested it with **24 HxBuddy submissions**, each changing one ingredient.

Income is not merit. Hours worked are, at exactly the committee's own weight. Distance and first-generation status add nothing.

Merit score tested	HxBuddy acc.
R + income	90.08
R only	92.28
R + income + hours + distance	92.38
R + hours + distance	94.38
R + 0.12 / 0.17 / 0.20 × hours	94.33 / 94.53 / 94.03
7 nearby weights and rates (0.135–0.158, 40.75–41.5%)	94.48 – 94.63
R + 0.146 × hours, 41%	**94.68**

audit_rapport.ipynb, sections 6 and 6b

METRIC CHOICE

Equal opportunity, measured against merit, not the committee

- **Why equal opportunity:** a merit award promises that a deserving student in Gaspé has the same chance as one in Montréal.
- **Why not demographic parity:** it forces equal rates whether or not merit differs, and says nothing about *who* gets the award.
- **The trap:** against the committee's labels the production model's TPR gap is only 0.06. It "passes" by copying the bias.

Merit reference (no penalty)	EO gap, production
Committee criteria, penalty removed	0.261
R score only	0.225
R score + hours worked	0.296
R + income + hours (linear)	0.236
Organisers' hidden reference	**0.270**

Our references bracket the published value: evidence the diagnosis is right.

audit_rapport.ipynb, sections 5 and 7

THE FIX

Make the model aware of the bias, then switch it off

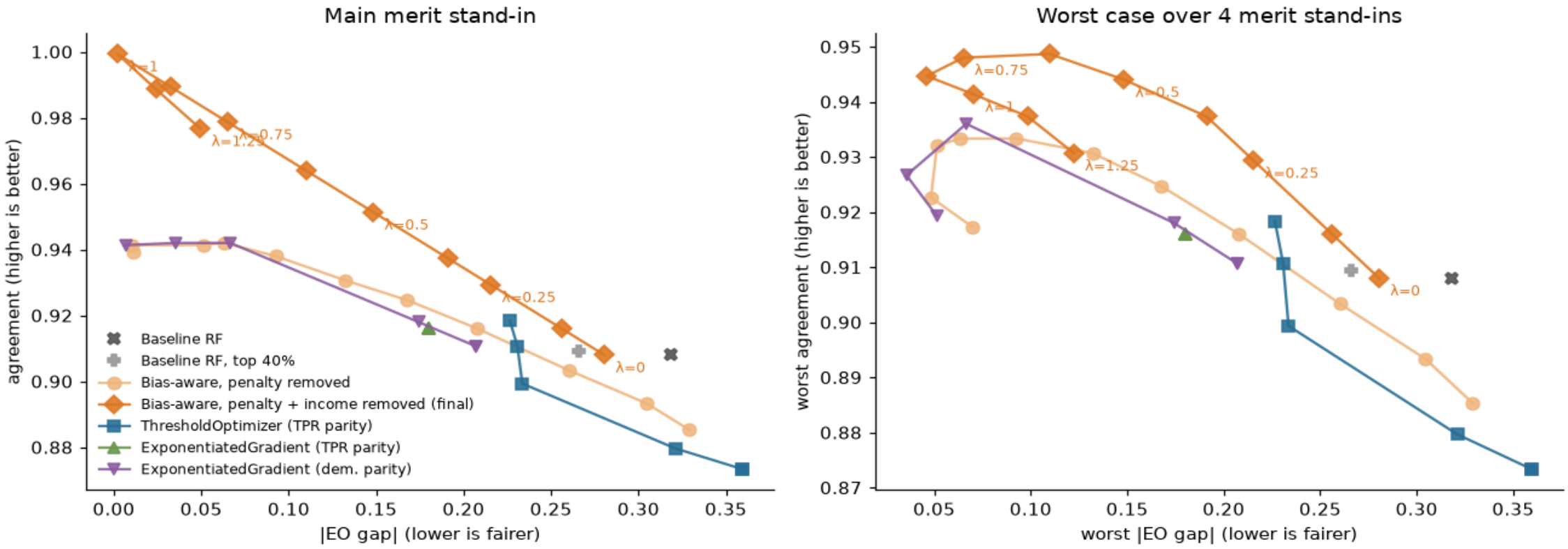
```
committee = 1.39·R + 0.20·hours + 1.78·log(income) + 0.11·distance − 2.17·remote
```

```
our score = R + 0.146 × weekly hours worked  → top 41% get an award
```

- **Train** with the remote indicator in the model, so it absorbs the penalty instead of the proxies.
- **Score** with the penalty, the income bonus and distance removed. Same merit, same score, wherever you live.
- **Budget**: 1,640 of 4,000 awards (41%), inside the 36–44% window.
- **Fairness knob**: λ, the share of the penalty removed. Sweeping it gives the Pareto front.

PARETO FRONT

Fairness and merit improve together



Held-out split, budget enforced. Dark diamonds: our final model. ThresholdOptimizer closes a third of the gap at best, and EG with TPR parity does not move: on biased labels the constraint is already met.

RESULTS ON THE 4,000 APPLICANTS

Gap closed, budget kept

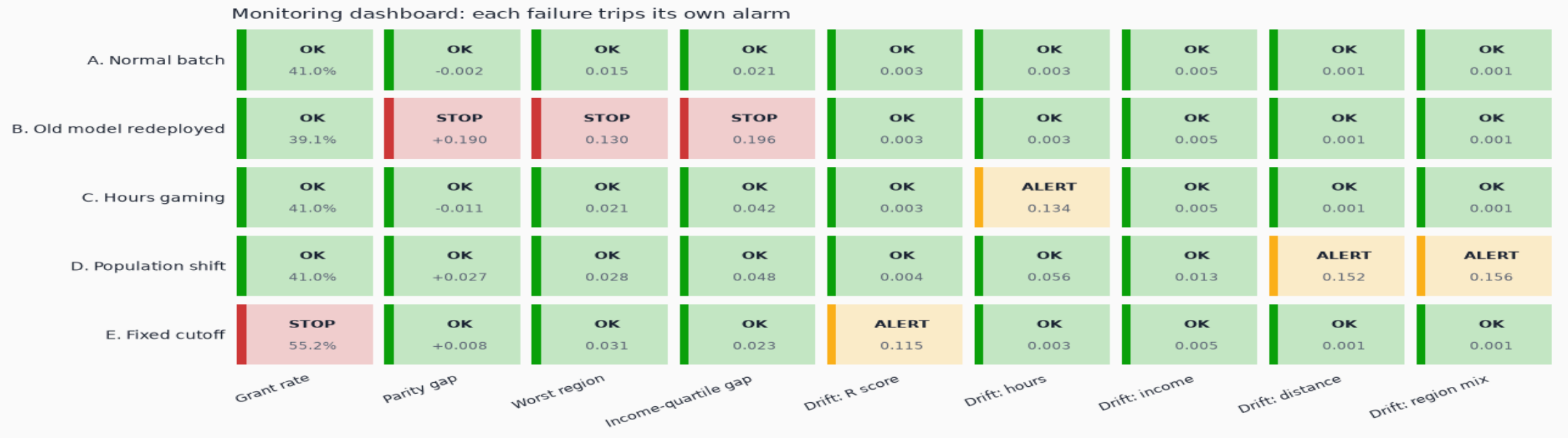
	Production	Ours
Grant rate	39.1%	**41.0%**
Grant rate centres / remote	46.8% / 27.8%	**40.9% / 41.1%**
EO gap vs R + hours merit reference	0.296	**0.000**
HxBuddy indicative accuracy / macro F1		**94.68% / 94.47%**
Accuracy vs past committee (held-out)	0.881	0.853

Same grades, different lives. Of 390 changed decisions, those who gain and those who lose have the same average R score (28.6). Those who gain work **twice the hours** (14.6 vs 7.4/week) with **half the family income** (45 k$ vs 84 k$).

model_corrige.ipynb, sections 8 and 8b · predictions.csv

GOVERNANCE

Monitoring is code, and we tested it on failures



- Old biased model redeployed: **STOP**, even though its budget looks normal. Hours inflated: **ALERT** (else 103 honest students lose out).
- Region and income only **measure** bias, never score. Law 25 notice and human review. Awards already granted are never withdrawn.

LIMITS

What we cannot claim

- We never saw the jury's reference. HxBuddy feedback chose two settings (drop income, 41%); three more rounds found nothing better, and we disclose all 24.
- If merit should ignore hours worked, our model slightly favours remote applicants (gap ≈ −0.05).
- ExponentiatedGradient with tight demographic parity is a close competitor. We prefer one readable formula.
- Synthetic data, two regional blocks. Other regions and a legal review are needed before roll-out.

Takeaway: the fix was not a better classifier. It was refusing to treat past decisions as the truth.

Tools: Python, pandas, scikit-learn, fairlearn, matplotlib · AI assistance: Claude Code (Anthropic)